Loan example - 50% LTV - 1 BTC, $100,000USD

Cadena Bitcoin

Loan Example Walkthrough

Loan Setup

- Borrower locks 1 BTC as collateral.
- Bitcoin price at the start: $100,000.
- 50% LTV → Loan notional size
  - 1 BTC * 50% = 0.5 BTC
- This 0.5 BTC amount represents $50,000 of value at inception.

Interest + Fee

- The loan has:
  - 10% annual interest
  - Cadena fee: 0.005 BTC
- Lender’s claim at maturity:
  - $50,000 USD worth of Bitcoin
- Cadena fee is add to Borrower’s contribution upfront

Borrower Proceeds

- Borrower receives the loan notional amount minus interest and fee upfront, because the lender's final claim is fixed at $50,000 worth of Bitcoin.
- So Borrower gets:
  - Loan notional (0.5 BTC)
  - Minus interest (0.05 BTC)
  - Minus Cadena fee (0.005 BTC)
- Borrower receives 0.445 BTC today.
- They still keep exposure to BTC because his 1 BTC collateral is still theirs—just locked in the DLC.

What the Borrower Gets

- Borrower gets:
  - Immediate liquidity (0.445 BTC worth of value)
  - Keeps full exposure to the Bitcoin price
  - No liquidation, no margin calls
  - At settlement he gets back whatever BTC is left after paying lender’s fixed claim
  - If BTC goes up → Borrower captures the upside
- This is basically:
  - Borrow cash without losing your Bitcoin.

What the Lender Gets

- Lender gets:
  - A guaranteed claim on $50,000 worth of BTC at maturity
  - No exposure to Bitcoin volatility
  - Their return is locked in Day 1
  - Cadena + DLC cryptography ensure their claim cannot be changed
- In fiat terms, lender earns 10% no matter what Bitcoin does.
  - Fixed, predictable return.

Settlement Logic

At the end of the year, the oracle reports the BTC price.
Lender gets exactly $50,000 worth of Bitcoin.
Borrower gets the rest of the 1 BTC.

If BTC rises

- Say BTC goes to $150,000.
- Lender gets 0.3333 BTC ($50,000)
- Borrower gets 1 - 0.3333 = 0.6667 BTC

If BTC stays the same

- BTC stays at $100,000.
- Lender gets 0.50 BTC ($50,000)
- Borrower gets 1 - 0.50 = 0.50 BTC

If BTC falls

- Say BTC goes to $70,000.
- Lender gets 0.7143 BTC ($55,000)
- Borrower gets 1 - 0.7143 = 0.2857 BTC

If BTC falls a lot

- Say BTC goes to $50,000 or below
- Lender gets 1 BTC ($50,000 or less)
- Borrower gets nothing

The Value Proposition

- Borrower benefits:
  - Keeps upside exposure to Bitcoin
  - No liquidations, no margin calls
  - Receives BTC liquidity upfront
  - Repayment happens automatically at settlement
  - Truly non-custodial

- Lender benefits:
  - Fixed, predictable return
  - Protected from BTC volatility
  - Trustless enforcement
  - No custody of collateral
  - Simple, safe, stable yield